

Crop Yield and Nitrous Oxide Emissions following Swine Manure Application: A Meta-Analysis

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Core Ideas

- Swine manure is a potential nutrient source but may harm environmental quality.
- Crops fertilized with swine manure matched those using synthetic fertilizer.
- Swine manure did not increase nitrous oxide compared with synthetic fertilizer.
- Manure application method and conditions dictate yield and environmental impacts.

Abstract: Swine manure has potential for reuse as fertilizer on cropland and pastureland. Currently, producers may be hesitant to land apply swine manure because information regarding the impacts to crop production and environmental quality has been inconsistent. The objective of this research was to synthesize the literature investigating impacts of swine manure application on crop yield and soil nitrous oxide emissions as compared to synthetic fertilizer. Overall, the meta-analysis of 39 studies shows that swine manure does not affect either crop yield or nitrous oxide emissions. However, a moderator analysis indicated that crop yield using manure fertilizer may be adversely affected in fine-textured soils or when applied to the soil surface. While these results suggest that swine manure may be safely applied as fertilizer, producers should still be cautious with how manure is applied to capture its N value without yield losses or detrimental effects to environmental quality.

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SWINE PRODUCTION is a major industry worldwide, contributing to the global economy and food security. In 2017, the United States produced over 72 million swine (USDA-NASS, 2019), resulting in an estimated 100 million Mg of manure (USEPA, 2013). Overall, this manure likely contains more than 0.85 million Mg nitrogen (N), 0.3 million Mg of phosphorus (P), and 0.48 million Mg of potassium (K) (Hatfield et al., 1998). Land application of manure utilizes these nutrients to meet crop demand, reduces the need for supplemental synthetic fertilizer, and removes the need for alternative disposal practices.

Swine manure application has the potential to increase crop yields (Wortman et al., 2017; Choudhary et al., 1996) but may also be associated with negative impacts to environmental quality, such as increased nutrient leaching (Choudhary et al., 1996) or increased ammonia emissions (Chantigny et al., 2007). Much of the uncertainty regarding swine manure application as fertilizer comes from divergent findings in the literature. For example, Schlegel et al. (2015) evaluated soil physiochemical properties after 10 yr of swine manure application and found that swine manure did not affect soil P, soil N, soil C, organic matter, or electrical conductivity. However, other evidence suggests that repeated manure applications increases soil nutrients and subsequent risk of losses through leaching, runoff, or gas emission (Ball-Coelho et al., 2004; Shepherd and Newell-Price, 2013).

Many of these inconsistencies stem from differences in site-specific characteristics, such as application rates, crop management systems, climate, and soil properties. In these circumstances, meta-analytic techniques may be used to evaluate the results of independent studies while accounting for a wide range of experimental conditions. The objective of this study was to conduct a meta-analysis to compare the effects of swine manure and synthetic fertilizer on both a measure of production, crop yield and a measure of environmental quality, nitrous oxide (N₂O) emissions.

Materials and Methods

Data Collection

A literature search was conducted using Web of Science (Clarivate) to find studies reporting a direct comparison between swine manure and synthetic N fertilizer in crop yield and/or N₂O emissions. Search terms included: “swine,” “pig,” “hog,” “manure,” “slurry,” “effluent,” “commercial,” “synthetic,” “mineral,” “urea,” “fertilizer,” “crop yield,” “fertilization,” and “production.” The search included only peer-reviewed articles published before 1 June 2018.

Studies were retained for analysis that met the following criteria: (i) replicated experimental design in field conditions, (ii) comparisons made within the same cropping and management system, (iii) yield for at least one full crop season (full season not required for studies reporting N₂O emissions), and (iv) no extreme conditions, such as flooding or major pest damage, occurred during the experiment. Additionally, the analysis included only studies with approximately the same amount of plant-available N supplied from the fertilizer. The mineralization factor of swine manure ranges from 0.3 to 0.5, depending on storage and handling conditions (Midwest Plan Service, 1993), and a conservative estimate for mineralization rate (30%) was assumed to allow for inclusion of more studies in the analysis.

Yield values from long-term studies (i.e., more than one season) were considered as a single observation if fertilizer practices remained constant and the average yield value from the study was reported. However, different application methods, crops, or soil sites reported within a single study were treated as independent observations. The list of studies included in the analyses is provided in the Supplemental Material.

Mean values for yield and/or N₂O emissions reported for swine manure and synthetic N fertilizer were extracted from each study, along with corresponding variance, sample size, and N application rate. GetData Graph Digitizer software (version 2.26) was used to extract any information are represented graphically in publications. N₂O emissions data included either (i) cumulative N₂O–N emissions for the study period (preferred) or (ii) average daily N₂O emission rates for the study period. Not enough studies were available to separate by length of experiment, so all N₂O emissions were analyzed together. Köppen climate classification, crop species, application method, soil pH, soil texture, and manure treatment information were also extracted for use in moderator analyses. To increase sample sizes, crop species were grouped into five levels: canola (*Brassica napus* L.), corn (*Zea mays* L.; both silage and grain), pasture (both grass and mixed), small grain (wheat [*Triticum aestivum* L.], oat [*Avena sativa* L.], barley [*Hordeum vulgare* L.], sorghum [*Sorghum bicolor* (L.) Moench]), and soybean [*Glycine max* (L.) Merr.]. Soil texture was grouped into three levels: coarse (sand, sandy loam), medium (loam, silt loam), and fine (clay loam, clay). Manure treatment was grouped into two levels: raw (no treatment) and treated (anaerobic digestion, composting, solid–liquid separation, filtration).

Data Analysis

The meta-analysis compared the impact of swine manure and synthetic N fertilizer on crop yield with the natural logarithm of the response ratio (ln RR) as the effect size of the comparison,

$$\ln \text{RR} = \ln \left(\frac{\bar{x}_1}{\bar{x}_2} \right) = \ln(\bar{x}_1) - \ln(\bar{x}_2) \quad [1]$$

where $\bar{x}_1$ and $\bar{x}_2$ are the mean values of swine manure application treatment and synthetic N fertilizer treatment, respectively. A random-effect meta-analysis model was applied assuming the effect sizes in the dataset are not intrinsically associated with each study k (Schwarzer et al., 2015). The model is shown in Eq. [2]:

$$\hat{\theta} = \theta + u_k + \sigma_k \varepsilon_k, \quad \varepsilon_k \sim N(0, 1); \quad u_k \sim N(0, \tau^2) \quad [2]$$

where $\hat{\theta}$ is the effect estimate from study k ; θ is the population effect; and u_k and $\sigma_k \varepsilon_k$ are the random effect and variance (with error) for each observation (Schwarzer et al., 2015). The random-effects model was applied separately for crop yield and N₂O emissions, as well as each moderator variable; all analyses were completed in R 3.5.1 software (R Core Team, 2018) using the ‘metafor’ package (Viechtbauer, 2010). The findings of the model were transformed to the percentage change of swine manure treatment compared with the synthetic N fertilizer treatment (Eq. [3]).

$$\% \text{ change} = [\exp(\hat{\theta}) - 1] \times 100 \quad [3]$$

Results And Discussion

Crop Yield

Across 84 pairwise comparisons in 34 studies, swine manure application reduced crop yields by 2.8% compared with synthetic N (Fig. 1) but was not statistically different ($p = 0.08$). This finding agrees with a review by Choudhary et al. (1996), who found that crops and pasture fertilized with swine manure matched or had slightly higher yields than those using inorganic fertilizer. The overall similarities in yield are likely due to crops receiving approximately the same amount of N. This finding confirms either that (i) the researchers accurately reported the amount of plant-available N in each study or (ii) using 30% as an estimate for plant-available N in the first season after application is suitable in a variable dataset. However, specific studies did report some inconsistencies with application rates targeting a specific plant-available N level, as researchers reporting decreased yield in canola determined that yields were diminished due to lower N availability from incorrect estimation of N mineralization (Vaillancourt et al., 2018).

While the overall effect of swine manure application on crop yield was not significant, the moderator analysis revealed some instances in which the swine manure significantly reduced yields (Fig. 1). Surface application can reduce corn yields by up to 14% (Mattila, 2006) because of increased N loss through ammonia volatilization, which reduces

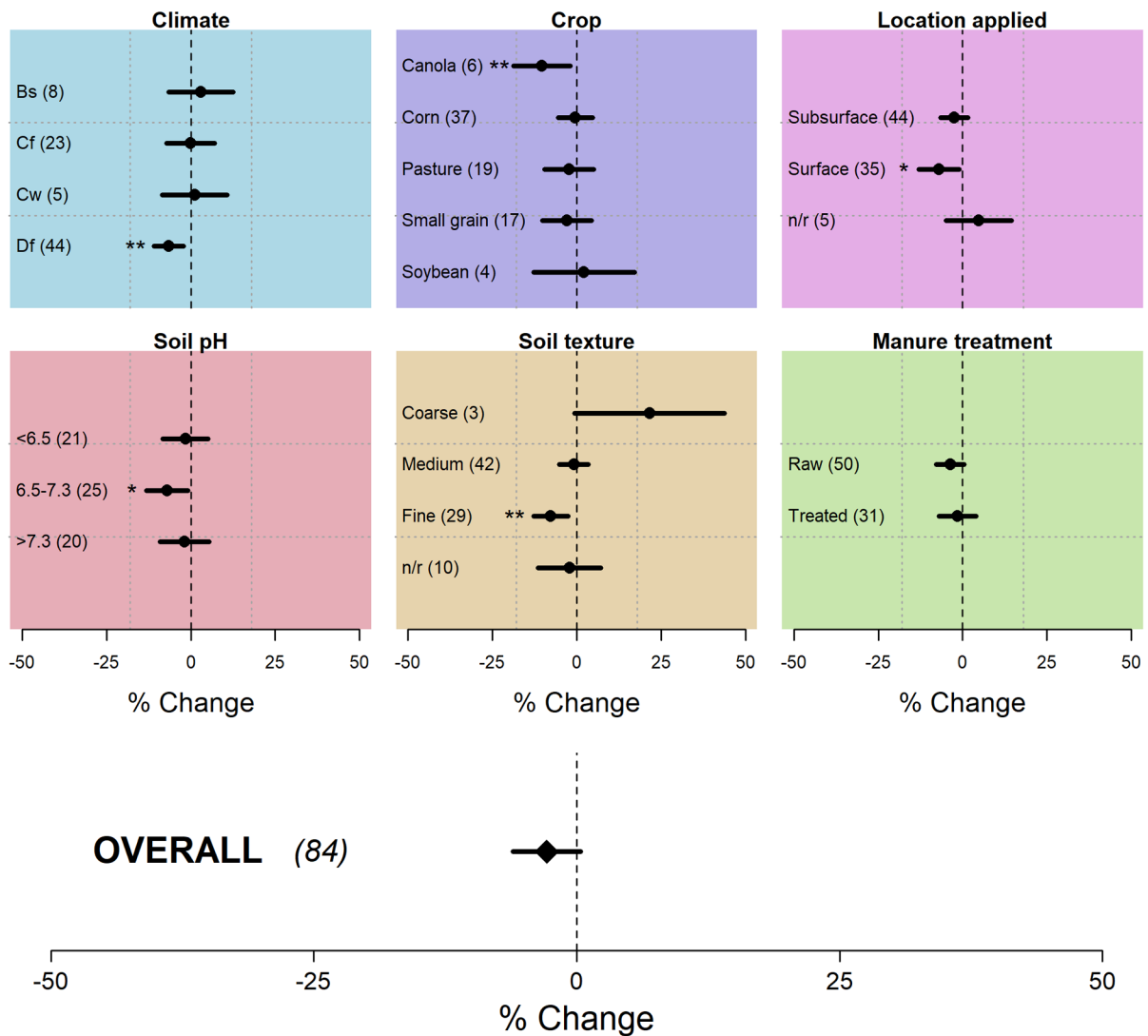


Fig 1. Forest plots from the moderator analyses (top) and the overall random-effects meta-analytic model (bottom pane) comparing swine manure fertilizer to synthetic N fertilizer on crop yield. Mean effect sizes are identified by points and 95% CI are identified by bars. The number of pairwise comparisons for each level included in the analyses is shown in parentheses. Results from the model confirm that bars not crossing the 0% line are significant at the $\alpha = 0.05$ level (*) or the $\alpha = 0.01$ level (**). n/r, study did not report information on moderator variable.

overall N agronomic efficiency (Gonzatto et al., 2017). The benefits from injection may be dependent on time of application, as Safley et al. (1980) reported that yield increases were higher in late winter than in fall or midwinter. Additionally, yield decreased in fine-textured soils, likely because they have higher redox potentials and are more susceptible to N_2O emissions, reducing the amount of N available for crop uptake (Rochette et al., 2008). However, this finding is not consistent in every circumstance, as swine manure increased yields in the clay soil compared with a loam in a direct comparison of two soil textures (Chantigny et al., 2008).

The reduced yield in Df climates (cold, humid continental climate; Beck et al., 2018) may be due both to the temperature dependence of N mineralization (Burger and Venterea, 2008) and to the practices in the region caused by a shorter growing season. The shorter growing season often leads producers in colder regions to apply larger amounts only one or two times per season. Conversely, regions with longer growing seasons may apply fertilizer, both manure and synthetic, more often at smaller rates, effectively reducing N losses

without a yield penalty, thereby improving N use efficiency (Huang et al., 2016).

Nitrous Oxide Emissions

Swine manure increased N_2O emissions by 3.5% (Fig. 2), but variability was so high that no differences were found ($p = 0.7$) in the dataset of 46 observations from 12 studies. Much of this variability comes from the inconsistency of measurement methods. Measurement periods ranged from 8 d to 2 yr and were collected using both periodic survey chamber samples and continuously collecting automated chambers. These temporal considerations are important, as N_2O emissions may be highest immediately following manure application but level out over time (Jarecki et al., 2008); thus, over an entire growing season, emissions may be the same or reduced as following synthetic fertilizer application (López-Fernández et al., 2007).

This variability in the dataset also outweighs some moderator effects that may be expected. For example, N_2O emissions would be expected to be higher in raw material than in

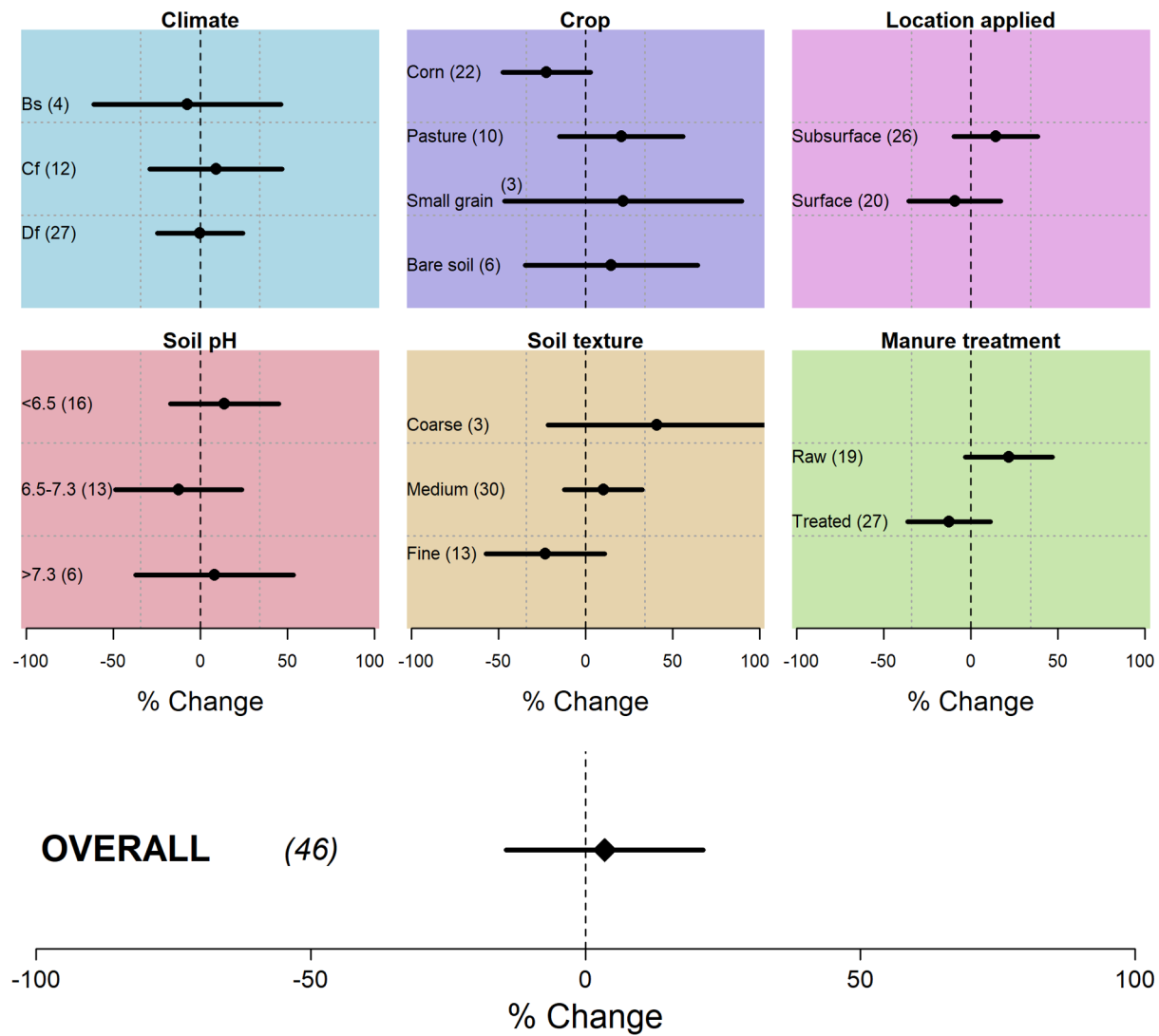


Fig 2. Forest plots from the moderator analyses (top) and the overall random-effects meta-analytic model (bottom pane) comparing swine manure fertilizer to synthetic N fertilizer on N_2O emissions. Mean effect sizes are identified by points and 95% CI are identified by bars. The number of pairwise comparisons for each level included in the analyses is shown in parentheses. Results from the model confirm that no moderator variables are not significant at the $\alpha = 0.05$ level, as no bars cross the 0% line.

treated (e.g., via anaerobic digestion or composting; Grave et al., 2018), as the purpose of treating the manure is to increase the nutrient use efficiency. Emissions are expected to be lower in the treated manure because most of the easily degradable C is decomposed, thereby reducing the possibility for denitrification (Chantigny et al., 2007). Conversely, subsurface (i.e., injection) of manure is expected to increase denitrification (Dell et al., 2011) by placing more C and N in contact, thereby increasing microbial activity and creating anaerobic conditions that favor denitrification (Wulf et al., 2002). The variability in this dataset suggests that either (i) these expectations may not hold in all circumstances or (ii) methods of measuring N_2O emissions are too variable to consistently identify these trends.

Implications

The findings of this meta-analysis indicate that swine manure can often be safely applied to cropland as a fertilizer without decreasing yield or increasing N_2O emissions compared with synthetic fertilizer. However, producers must

be cognizant of site-specific conditions to determine how best to capture the nutrients contained in the manure. For example, producers working in humid temperate climates or with fine-textured soils may suffer slight yield declines, likely due to low mineralization rates. These producers must then weigh those yield declines against the cost savings from removing the need for alternative manure disposal and reducing the amount of synthetic fertilizer needed to be purchased. Similarly, a trade-off may exist between surface and subsurface application, where surface application may diminish yield but subsurface application may increase N_2O emissions. Overall, crop yield and N_2O emissions are valuable in assessing the impacts of swine manure to cropland, but manure application affects soils in additional ways, such as increasing soil organic matter and creating more ammonia emissions, especially when applied on the soil surface. Meta-analysis can be a useful tool in identifying the effects of management strategies, but it is only as powerful as the body of literature it describes. The findings of this meta-analysis would be more robust, for example, with more studies on

coarse-textured soils or in temperate dry climates. Similarly, studies reporting N_2O emissions over the same time period and same length of time following application could reduce variability in the meta-analytic model and provide more meaningful results. Thus, future research can better quantify the value of applying swine manure by (i) taking a more holistic approach and including more parameters in swine manure application research and (ii) standardizing methodology to allow for more meaningful comparisons between studies.

Supplemental Material

Included in the supplemental material are two reference lists. Reference List 1 includes all studies included in the crop production meta-analysis, and Reference List 2 includes all studies included in the N_2O meta-analysis. Studies used in both analyses are identified in Reference List 1.

Conflict of Interest

The authors declare no conflict of interest.

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